

IWOTO, Japan

WMO: 479810

Lat: 24.784N

Lon: 141.322E

Elev: 394

StdP: 14.49

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	60.6	61.3	43.1	41.6	62.6	45.0	44.8	63.4	26.9	69.0	24.9	68.8	12.8	20	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	87.8	79.4	86.4	78.9	86.1	78.7	80.9	84.5	80.2	84.0	79.7	83.6	13.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	157.0	83.3	79.1	153.1	82.9	78.8	151.6	82.8	45.0	85.5	44.4	84.7	43.9	82.8	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.6	23.6	21.3	DB	58.9	89.0	1.4	0.9	57.9	89.6	57.1	90.1	56.2	90.6	55.2	91.2	
			WB	50.2	81.7	1.5	1.0	49.1	82.4	48.3	83.0	47.4	83.6	46.4	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	67.7	67.7	69.3	72.7	76.4	80.7	82.4	82.2	82.0	80.7	76.9	71.5
	DBStd	6.16	2.68	3.17	3.15	3.10	2.83	2.48	1.68	1.76	1.76	1.79	2.77	3.47
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	20	6	10	3	0	0	0	0	0	0	0	0	1
	CDD50	9448	547	495	599	681	817	920	1005	999	959	951	807	668
	CDD65	3993	89	84	137	231	352	470	540	534	509	486	357	204
	CDH74	31542	26	43	130	582	1941	4570	5974	5794	5390	4613	2088	391
	CDH80	6979	0	0	0	12	151	1058	1743	1593	1391	855	171	5
(14) Wind	WSAvg	12.4	12.7	12.8	13.2	12.7	11.9	10.4	10.9	12.6	12.7	12.2	13.1	13.1
(15) Precipitation	PrecAvg	47.9	2.0	1.5	1.9	3.0	6.1	2.9	5.2	8.1	5.9	4.4	4.2	2.7
	PrecMax	71.1	4.2	2.5	4.7	7.8	14.7	11.8	13.3	23.8	13.1	15.4	9.7	6.1
	PrecMin	28.8	0.5	0.2	0.4	0.3	1.0	0.0	0.5	1.3	0.5	0.5	0.9	0.8
	PrecStd	10.7	0.9	0.6	1.1	2.1	3.8	2.3	3.7	5.3	3.4	3.7	2.6	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	77.2	78.8	81.0	84.3	87.9	88.2	88.2	88.1	86.4	84.5	80.7
		MCWB	70.9	71.0	73.1	75.4	78.1	79.2	79.1	79.6	79.7	79.1	77.4	74.8
	2%	DB	74.9	75.3	77.0	79.3	82.7	86.3	87.7	87.7	86.4	85.9	82.8	78.9
		MCWB	68.9	69.5	71.9	74.2	76.9	78.6	79.1	79.6	79.1	78.9	76.2	73.3
	5%	DB	73.3	73.6	75.3	78.7	82.0	85.8	86.3	86.3	86.1	84.5	82.2	77.3
		MCWB	67.3	68.2	70.3	73.8	76.5	78.5	78.6	78.9	78.9	78.0	75.8	72.0
	10%	DB	71.8	72.0	73.7	77.1	80.6	84.4	85.9	85.8	85.6	84.1	80.7	76.0
		MCWB	65.3	66.5	69.1	73.1	75.8	77.8	78.5	78.7	78.6	77.8	74.7	71.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	72.7	74.5	77.2	79.2	81.2	81.4	81.7	81.2	81.1	79.1	76.5
		MCDB	75.0	75.2	77.1	79.9	82.1	84.4	85.1	85.0	84.5	84.1	82.2	78.7
	2%	WB	70.3	71.1	72.9	76.0	78.3	80.1	80.4	80.7	80.5	79.9	78.1	75.2
		MCDB	73.2	73.9	75.5	78.5	81.2	83.6	84.7	84.5	84.4	83.2	81.3	77.4
	5%	WB	68.7	69.8	71.9	75.0	77.6	79.2	79.7	79.9	79.8	79.1	77.2	74.0
		MCDB	72.1	72.8	74.4	77.2	80.4	83.0	84.1	83.9	83.8	82.6	80.5	76.5
	10%	WB	67.0	68.3	70.6	74.1	76.7	78.8	79.1	79.2	79.2	78.7	76.2	72.4
		MCDB	70.9	71.7	73.3	76.3	79.4	82.7	83.6	83.3	83.2	82.2	79.7	75.5
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	6.9	6.7	6.3	6.5	6.7	6.5	6.4	6.2	5.7	5.8
		MCDBR	7.4	7.7	7.9	7.6	7.1	7.2	7.3	7.3	7.0	6.7	6.3	6.5
	5% WB	MCWBR	6.4	6.0	6.0	4.9	4.2	3.5	3.6	3.6	3.6	3.8	4.1	5.1
		MCWBR	7.3	7.3	7.4	6.9	6.6	6.7	6.9	6.6	6.6	6.3	5.9	6.3
(40) Clear-Sky Solar Irradiance	taub		0.379	0.392	0.440	0.461	0.437	0.403	0.417	0.438	0.429	0.418	0.403	0.399
		taud	2.472	2.428	2.292	2.282	2.431	2.547	2.504	2.444	2.464	2.488	2.506	2.444
	Ebn at Noon		278	282	273	269	273	280	277	272	273	271	268	266
		Edh at Noon	31	35	42	43	37	33	34	36	35	33	30	31
(44) All-Sky Solar Radiation	RadAvg		1140	1416	1631	1831	1884	2171	2107	1931	1823	1582	1251	998
	RadStd		87	96	117	169	181	163	142	100	108	100	92	102

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.58	+0.70	+0.83	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page